Benchmarking and Advancing Generative Models for Calorimeter Simulation

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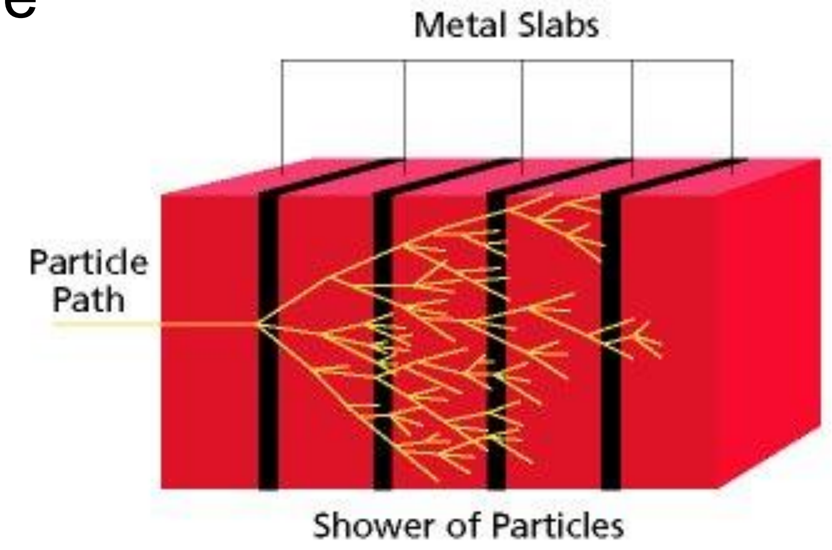
Outline

- Background
- Motivation & Research Question
- Hypotheses
- Contributions
- Related work & Dataset
- RQ1: Proposed evaluation metric
- RQ2: Framework for benchmark
- RQ3: Application of physics feature in generative model.
- Discussion
- Conclusion & Future work

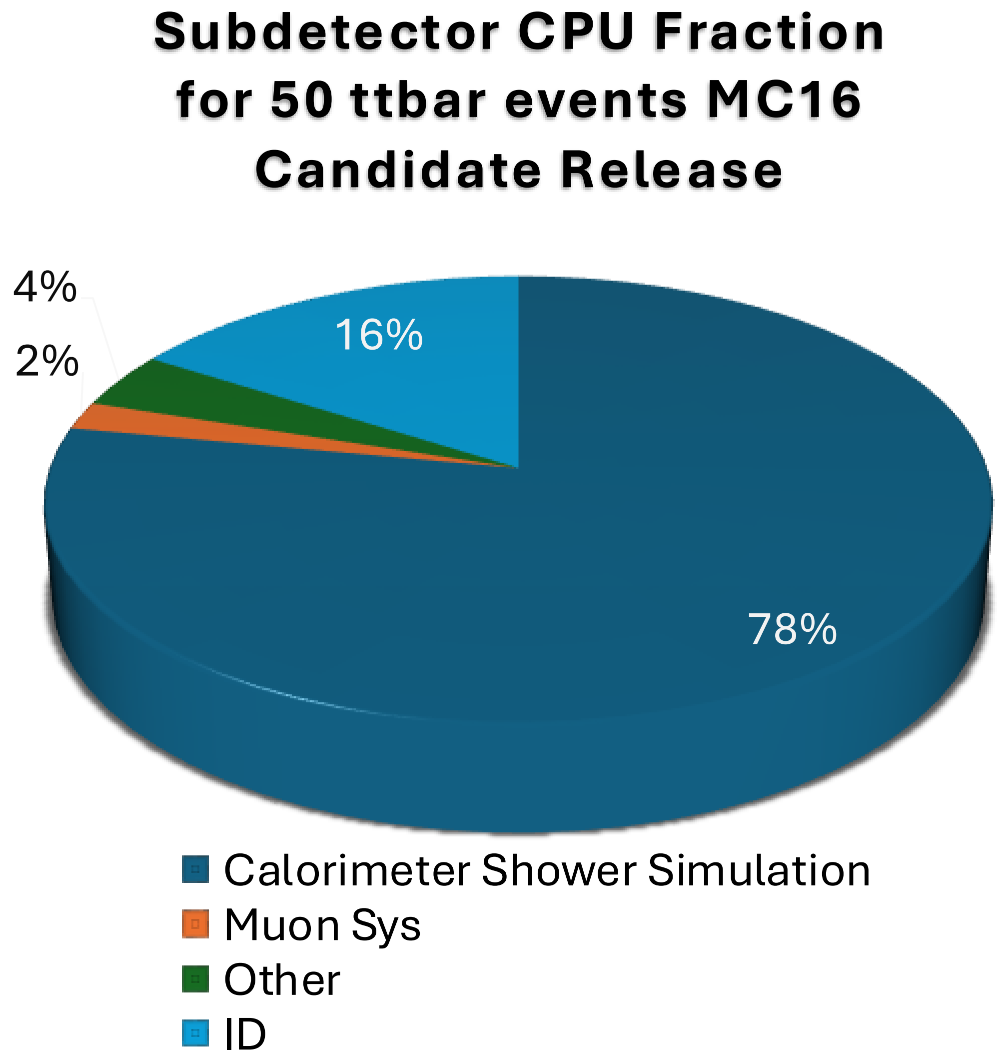
Background: Calorimeters

- The calorimeter is a detector used to measure the energy of particles
- When high-energy particles enter the calorimeter, they produce what we call a particle shower.
- This shower is essentially a cascade of secondary particles generated as the initial high-energy particle interacts with the dense matter inside the calorimeter.
- Crucial for reconstructing particle energies and identifying particles at the LHC.



Background: Geant4

- Geant4[12] is a Monte Carlo toolkit for simulating particle interactions in detectors
- ✓ Delivers high-fidelity shower simulations but is computationally intensive
- x Calorimeter simulation accounts for >75 % of total simulation time[13]



Background: Fast Simulation Alternatives

- HL-LHC will dramatically increase luminosity after 2029, accumulating more data and requiring higher-precision simulations
- Geant4 cannot scale to the required number of events — simulation becomes the bottleneck
- Parametric fast simulators (FastCaloSim/V2)[14] reduce time via manual tuning to Geant4 but oversimplify phenomena and require calibration.

Motivation & Research Questions

Motivation

- Existing metrics (from physics, computer vision, and statistics) focus mainly on high-level shower features.
- They often **fail to capture spatial correlations** across layers and voxels, leaving key physics-relevant dependencies underexplored.

Research Question 1(RQ1)

- How can we design and evaluate generative models that capture high-dimensional correlation structures in scientific data, and how effective is Correlation Frobenius Distance (CFD) as a metric for quantifying such dependencies?

Motivation & Research Questions

Motivation

- Lots of Proposed Generative Model but **no systematic framework to evaluate**.
- Each model has strengths and weaknesses.

Research Question 2(RQ2)

- What are the comparative strengths and weaknesses of GAN-, VAE-,Flow-, and Diffusion-based generative models for calorimeter shower simulation when evaluated with physics-informed, statistical, and computer vision metrics?

Motivation & Research Questions

Motivation

- Since the Correlation Frobenius Distance (CFD) analysis indicates that the deposited energies in calorimeter layers and voxels are correlated.

Research Question 3(RQ3)

- Can a two-stage autoregressive diffusion model (Lantern) improve correlation fidelity and flexibility compared to existing generative paradigms?

Hypotheses

Hypothesis 1

- Incorporating domain-informed correlation structures into loss functions or evaluation metrics will improve the quality of generative models and highlight discrepancies that traditional statistical or computer vision metrics fail to capture.

Hypothesis 2

- *A factorized, two-stage autoregressive diffusion model* ***'Lantern'*** *can better capture both layer-wise and voxel-level correlations than existing state-of-the-art generative approaches.*

Contributions

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Correlation analysis with CFD

Introduced Correlation Frobenius Distance (CFD) for layer-wise and voxel-wise correlations.

Provides a physics-sensitive evaluation beyond PCC and computer vision metrics.

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Systematic Evaluation Framework

Benchmarked GAN, VAE, Flow, and Diffusion models

Compared strengths and weaknesses with physics-informed, statistical, and CV metrics

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Compared strengths and weaknesses with physics-informed, statistical, and CV metrics

Lantern Model

Two-stage autoregressive diffusion model (energy prediction + voxel generation)

Captures correlations with fewer parameters .

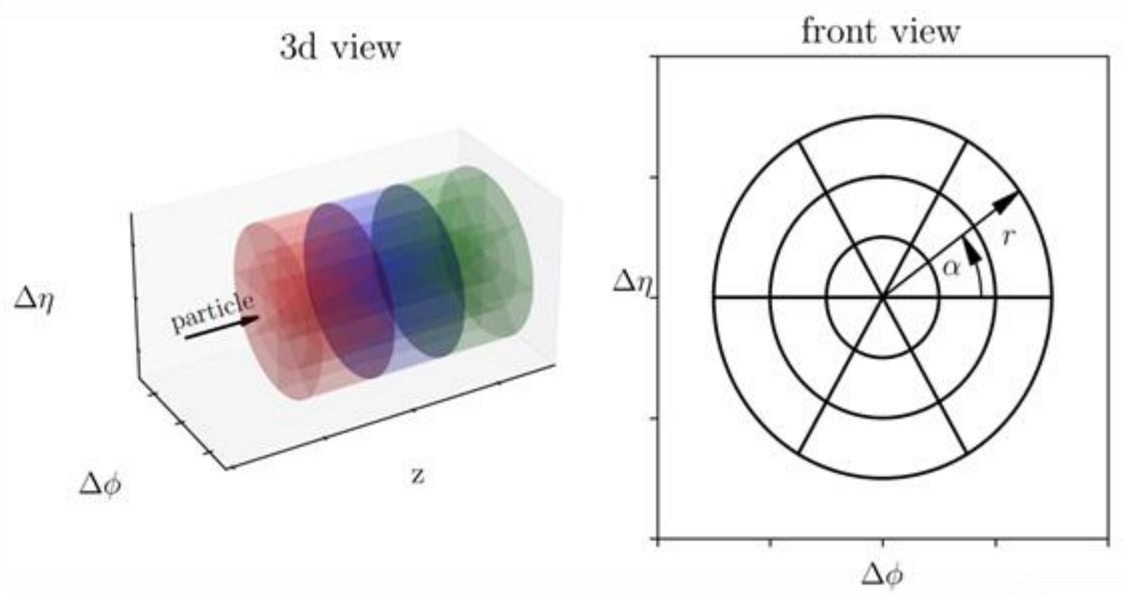
Related Work

Model	Speed	Fidelity	Challenges
Generative Adversarial Network[15,16,17]	Fast	Moderate	Unstable training, Mode collapse
Variational AutoEncoder[18,19,20]	Very fast	Low	Blurry samples, less detailed
Normalizing Flow[6,21,22,23]	Moderate	High	Restrictive bijective mapping
Diffusion[2,4,5]	Slow	Highest	Slow sampling, computational cost
Flow Matching[3,24]	Slow	Highest	Slow sampling

Datasets

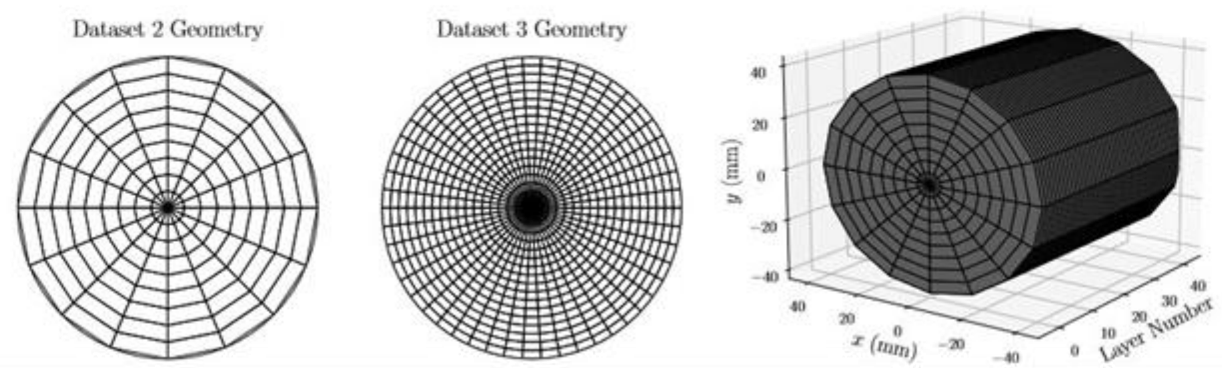
- **Three benchmark datasets** from CaloChallenge 2022[11], designed to test generative models for calorimeter shower simulation.
- Difficulty levels defined as *easy, medium, hard* based on **voxel dimensionality and granularity**.
- Common **cylindrical calorimeter geometry** inspired by the ATLAS detector at the LHC.
- Particles propagate along the **z-axis**, with energy depositions recorded across discrete layers.
- Captures both **longitudinal (layer-wise)** and **transverse (radial/ angular)** shower development → a high-dimensional challenge for generative models.

Datasets



CaloChallenge datasets scale in complexity: from low-dimensional photon/pion showers (~hundreds of voxels) to high-granularity electron showers (~40k voxels), posing increasing challenges for generative models.

Calorimeter showers are recorded as energy deposited across layers, segmented radially and angularly, with particles propagating along the z-axis.



Datasets

Dataset	Particle Types	Voxels per Shower	Layers	Segmentation ($N\alpha \times Nr$)	Energy Range
1	Photons, Pions	368 (γ), 533 (π^+)	5–7	Varies by layer	256 MeV–4 TeV
2	Electrons	6,480	45	16 angular × 9 radial per layer	1–1000 GeV
3	Electrons	40,500	45	50 angular × 18 radial per layer	1–1000 GeV

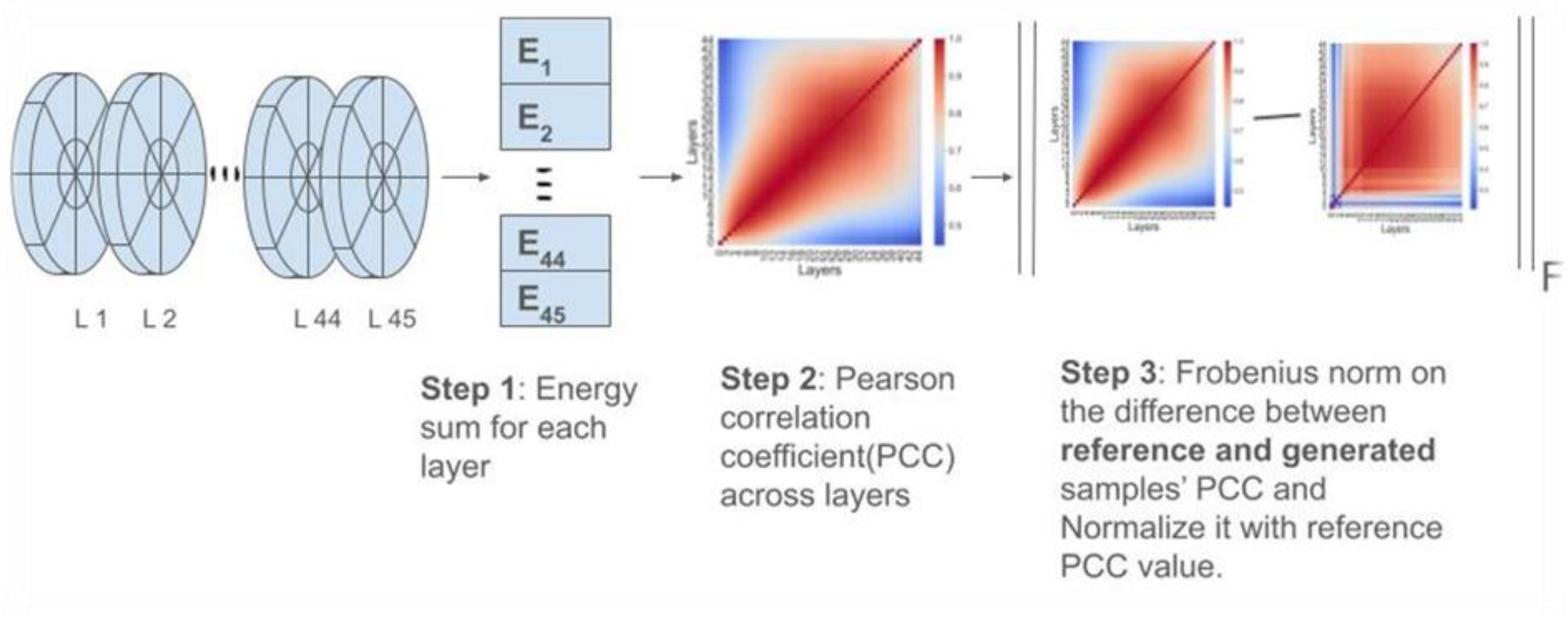
RQ1: Correlation Frobenius Distance

- How can we design and evaluate generative models that capture high-dimensional correlation structures in scientific data, and how effective is Correlation Frobenius Distance (CFD) as a metric for quantifying such dependencies?

- Correlation Frobenius Distance (CFD)
  - Applied against Geant4 reference samples; ideal value = **0**
  - Lower CFD → better model captures correlations.
  - **Layer-wise CFD:** Measures differences in correlations between energy deposits across calorimeter layers.
  - **Voxel-wise CFD:** Measures differences in local spatial correlations between voxels in consecutive layers.

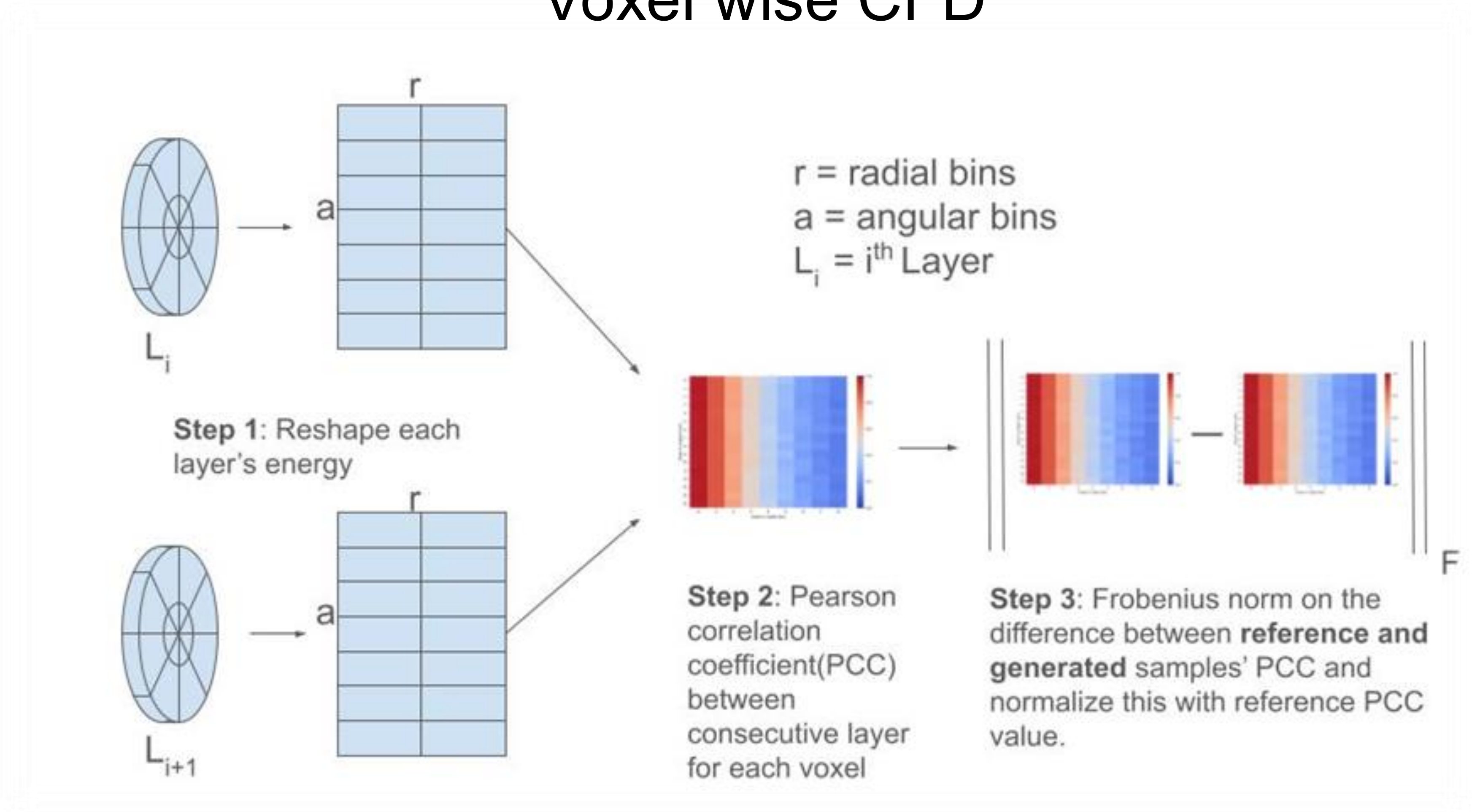
RQ1: Correlation Frobenius Distance

Layer wise CFD



RQ1: Correlation Frobenius Distance

Voxel wise CFD

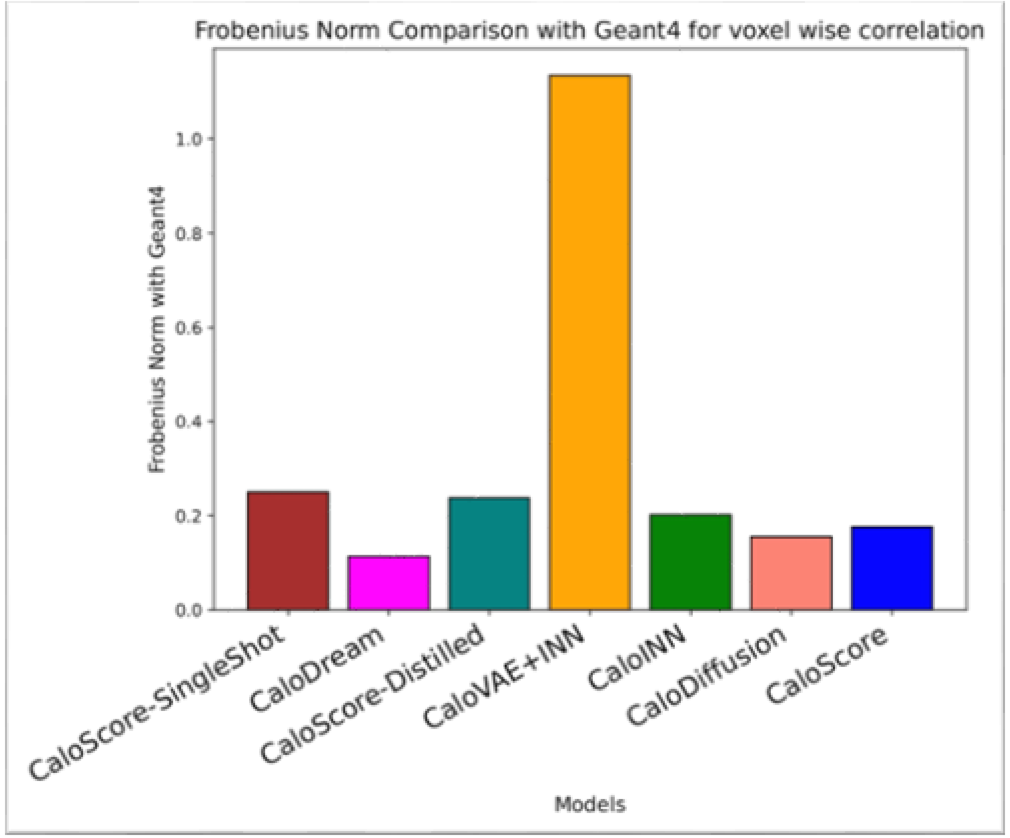


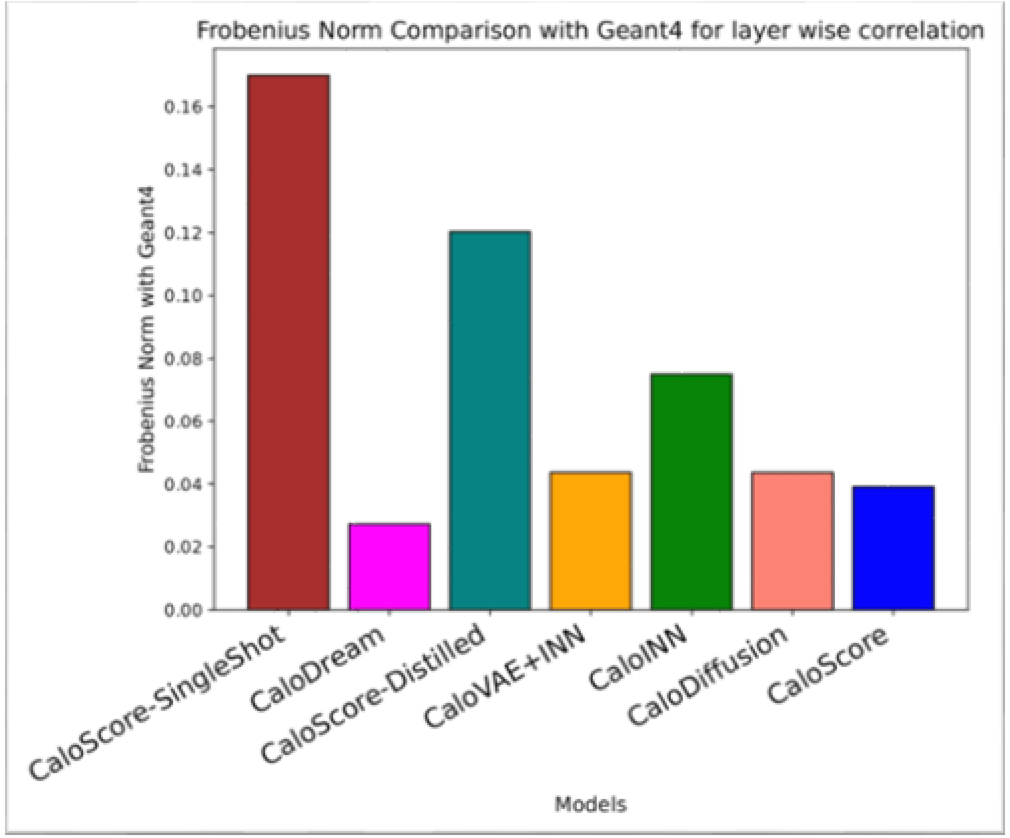
RQ1: Correlation Frobenius Distance

Model Name	Gen AI approach
CaloDiffusion[2]	Diffusion Model[7]
CaloDream[3]	Flow Matching[8]
CaloScore[4,5] CaloScore Distilled, CaloScore Single Shot	Score Based Model[9]
CaloINN[6] CaloVAE+INN	VAE+Normalizing Flow[10]

RQ1: Correlation Frobenius Distance

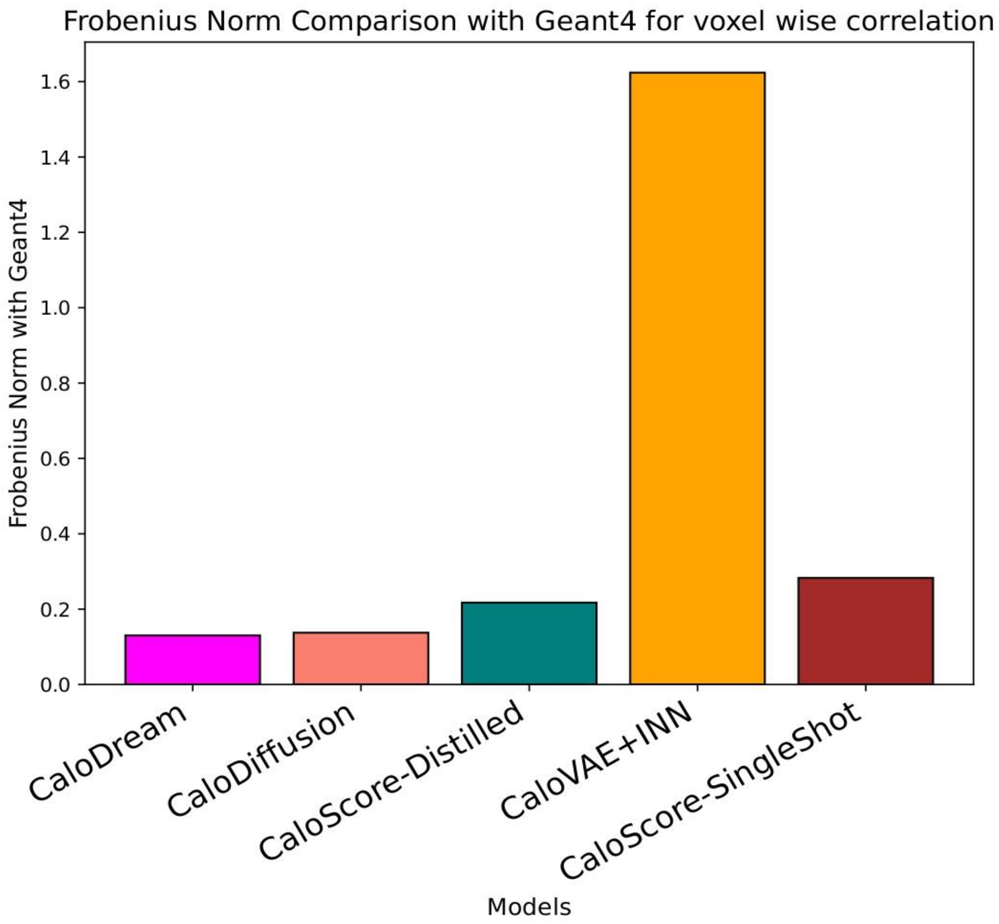
Results on Dataset 2

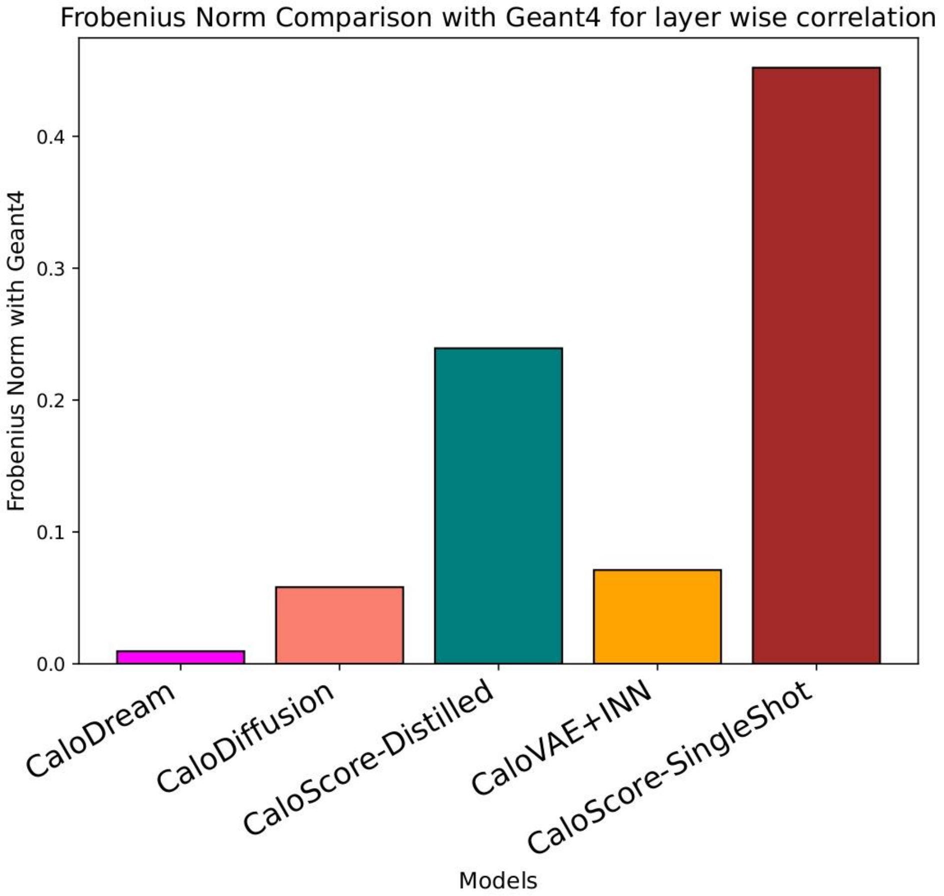




RQ1: Correlation Frobenius Distance

Results on Dataset 3





RQ2

- What are the comparative strengths and weaknesses of GAN-, VAE-,Flow-, and Diffusion-based generative models for calorimeter shower simulation when evaluated with physics-informed, statistical, and computer vision metrics?

Model Name	Gen AI approach
CaloDiffusion[2]	Diffusion Model[7]
CaloDream[3]	Flow Matching[8]
CaloScore[4,5]	Score Based Model[9]
CaloINN[6]	VAE+Normalizing Flow[10]

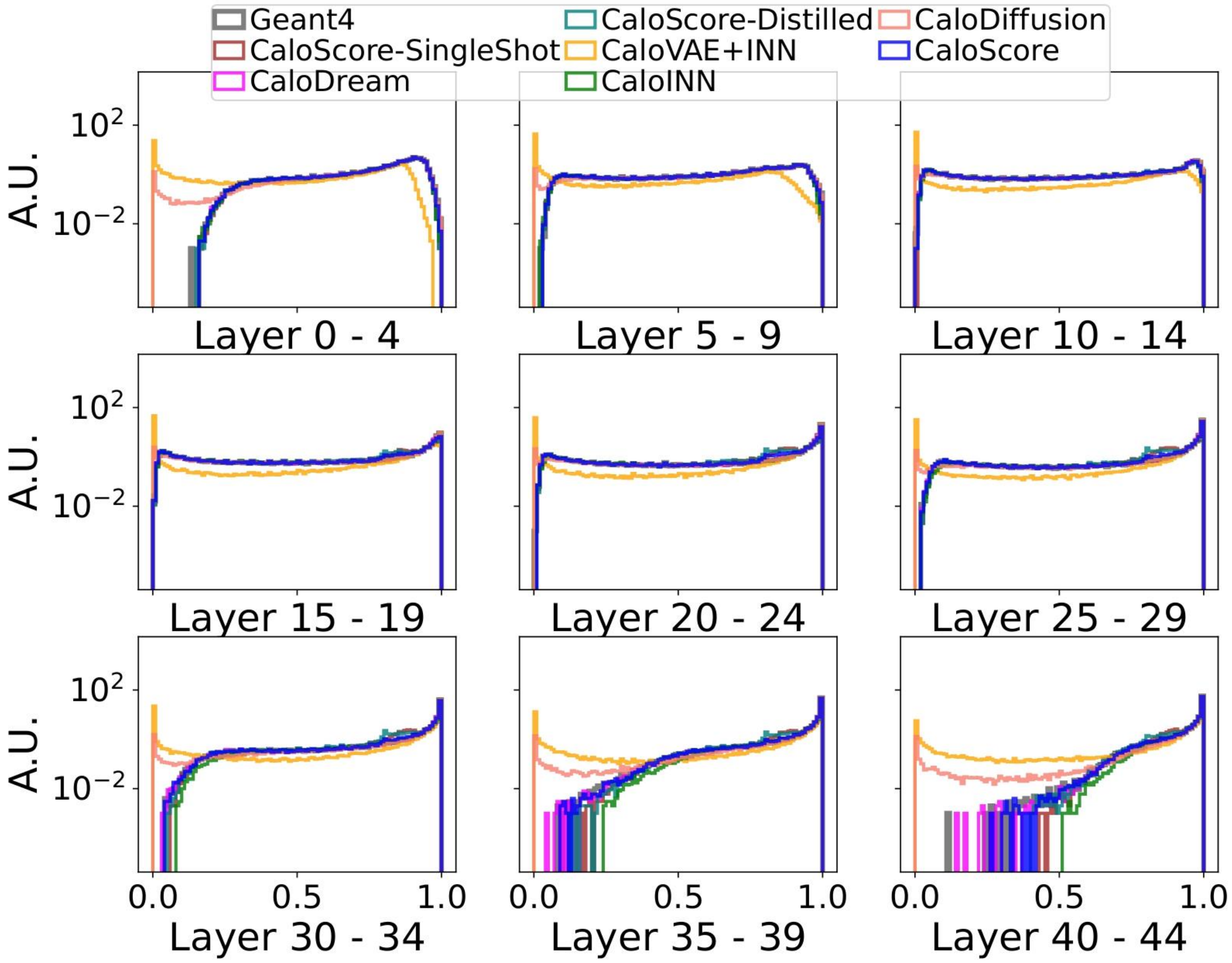
Our study reveals that Flow Matching based model performs better with respect to sample quality. Second comes the Diffusion based model.

RQ2: Evaluation Metrics[1]

Category	Metrics	What They Measure
Physics Observables	Layer-wise energy, Total energy, Center & Width (η, φ), Sparsity	High-level shower distributions
Distributional Distances	Separation Power, Earth Mover's Distance (EMD)	Histogram / 1D distribution similarity
Classifier-based	Classifier Test (AUC)[21]	Ability to distinguish Geant4 vs. generated showers
Manifold-based	Precision, Recall, Density, Coverage (PRDC)[26,27]	Quality, diversity, support coverage
Physics-aware	Fréchet Physics Distance (FPD), Kernel Physics Distance (KPD)[25]	Multivariate physics feature similarity
Correlation-based	Pearson Correlation Coefficient (PCC), Correlation Frobenius Distance (CFD)	Layer-wise & voxel-wise correlations
Model Complexity	Number of parameters	Computational cost / model size
Timing	Generation time vs. Geant4	Efficiency & speed-up

RQ2: Results (Dataset 2)

- Sparsity modeling is critical for realistic detector response and physics tasks.
- CaloScore, CaloINN, and CaloDREAM match Geant4 well, while CaloDiffusion and CaloVAE+INN struggle, leading to biased energy reconstruction.



RQ2: Results (Dataset 2)

Model	FPD/KPD (↓)	Classifier Test (~0.5) (Low and High level features)	Precision, Recall, Density, Coverage (PRDC) (↑)	# of Parameters (↓)
CaloDiffusion	moderate	≈ 0.5-0.6	balanced	~0.52M
CaloDream	low	~0.5	balanced	~28M
CaloScore	moderate	0.55-0.7	balanced	~14M
CaloINN	high	>0.7	imbalanced	~271M

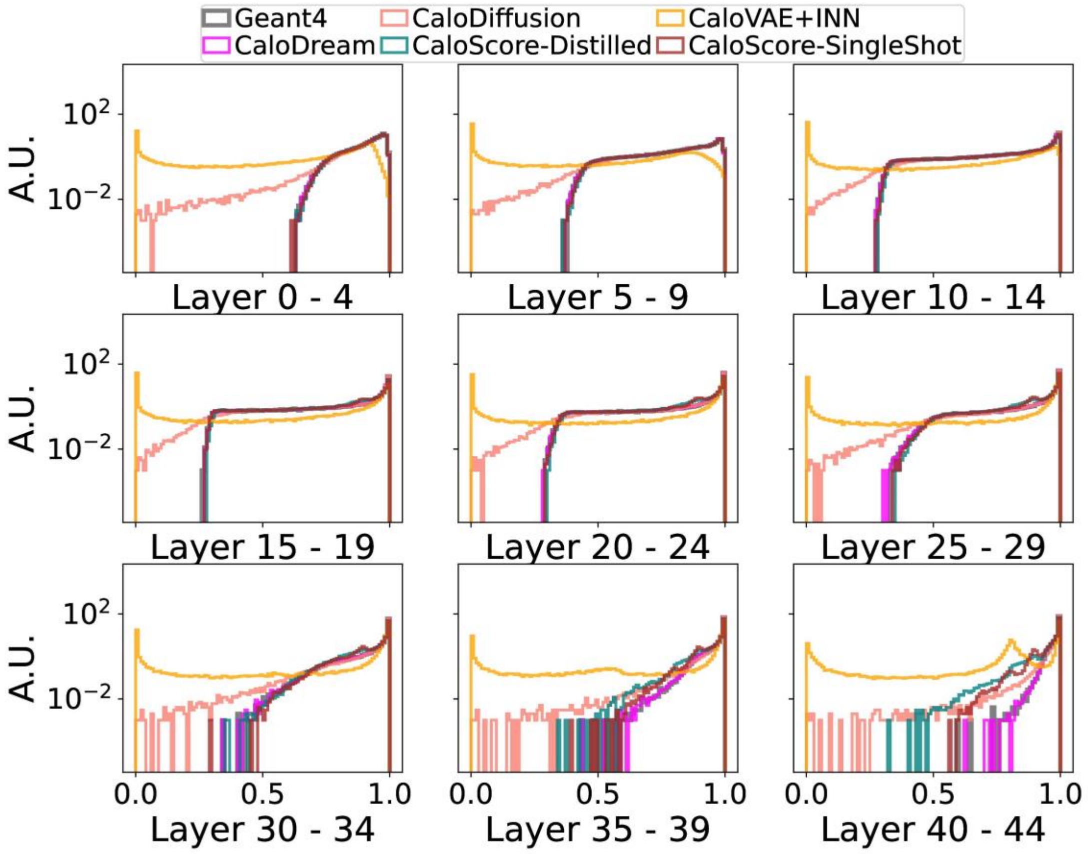
RQ2: Results (Dataset 2)

Model	Accuracy	Generation Speed	Memory Requirement
CaloDiffusion	Moderate	Moderate	Low
CaloDream	High	Moderate	Moderate
CaloScore	Moderate	Slowest	Moderate
CaloINN	Low	Fastest	High

RQ2: Results (Dataset 3)

- Applied to **CaloDiffusion, CaloDream, CaloVAE+INN, and CaloScore variants (distilled & single-shot)** due to high-dimensional voxel data.

- **CaloDream outperforms others** in capturing sparsity, while CaloDiffusion and CaloScore distilled lag behind.



RQ2: Results (Dataset 3)

Model	FPD/KPD (↓)	Classifier Test (~0.5) (Low and High level features)	Precision, Recall, Density, Coverage (PRDC) (↑)	# of Parameters (↓)
CaloDiffusion	moderate	≈ 0.56-0.60	balanced	~1.2M
CaloDream	low	~0.50-0.65	balanced	~8.2M
CaloScore Distilled	moderate	>0.7	imbalanced	~29M
CaloVAE+INN	high	>0.8	imbalanced	~204M

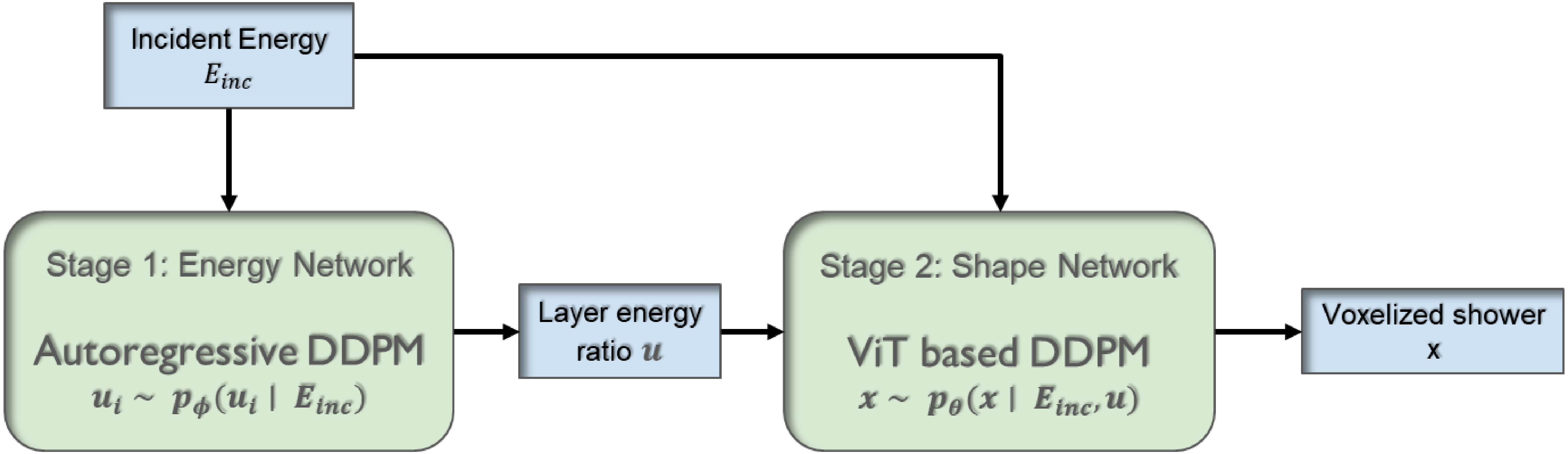
RQ2: Results (Dataset 3)

Model	Accuracy	Generation Speed	Memory Requirement
CaloDiffusion	Moderate	Moderate	Low
CaloDream	High	Moderate	Moderate
CaloScore Distilled	Moderate	Slowest	Moderate
CaloVAE+INN	Low	Fastest	High

RQ3: Lantern

- Can a two-stage autoregressive diffusion model (Lantern) improve correlation fidelity and flexibility compared to existing generative paradigms?
  - We propose ***Lantern***, **a two-stage autoregressive diffusion model** for calorimeter simulation. In the first stage, **an autoregressive DDPM predicts layer-wise energy sums;** in the second, a **Vision Transformer–based DDPM generates voxel-level showers conditioned on these energies.**
  - We evaluated Lantern on **Dataset 2** of CaloChallenge 2022.

RQ3: Lantern



Incident Energy
E_{inc}
Stage 1: Energy Network
Autoregressive DDPM
$u_i \sim p_\phi(u_i \mid E_{inc})$
Layer energy ratio u
Stage 2: Shape Network
ViT based DDPM
$x \sim p_\theta(x \mid E_{inc}, u)$
Voxelized shower x

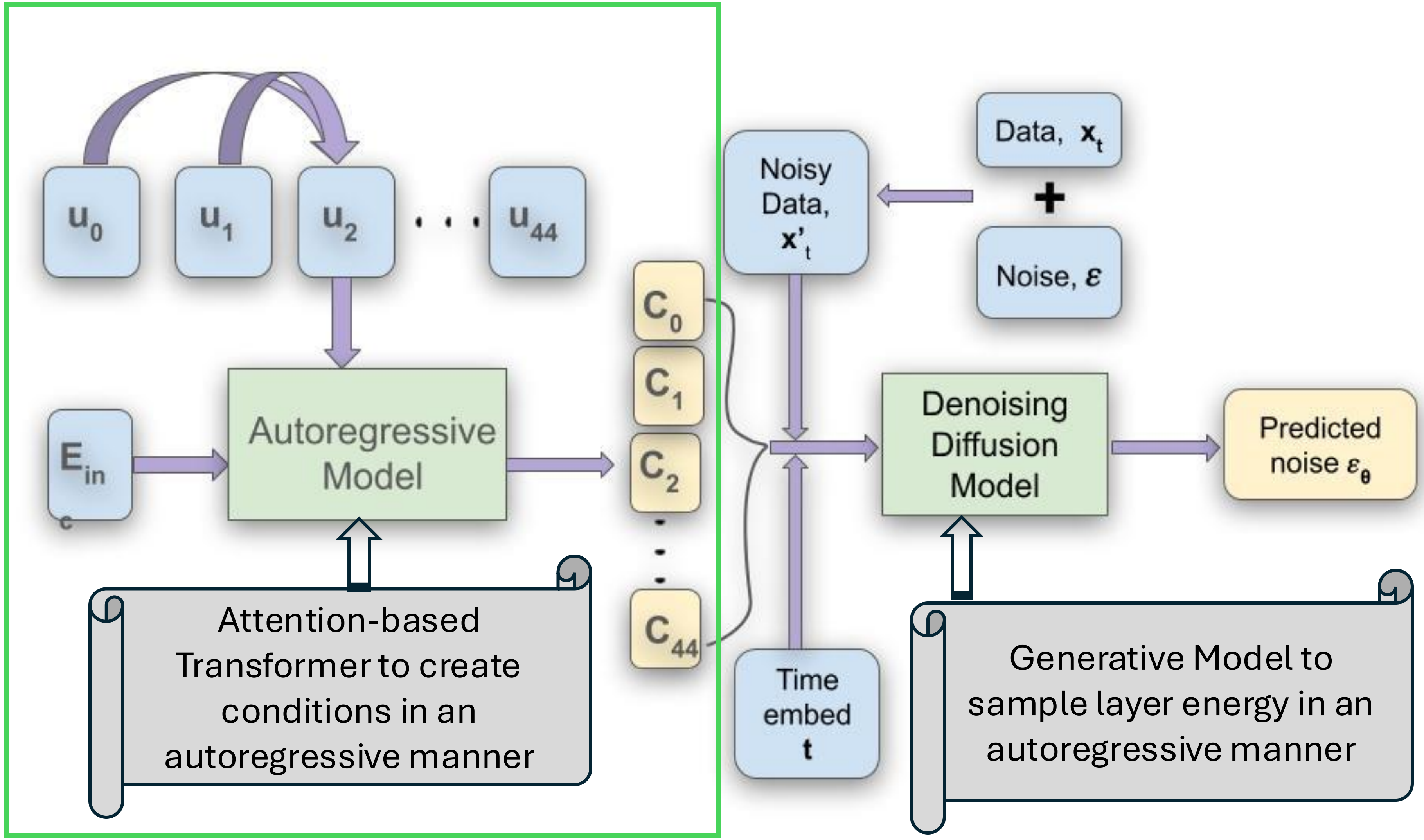
RQ3: Energy Network Preprocessing

- Introduce **energy ratios** to encode deposited energy across layers:

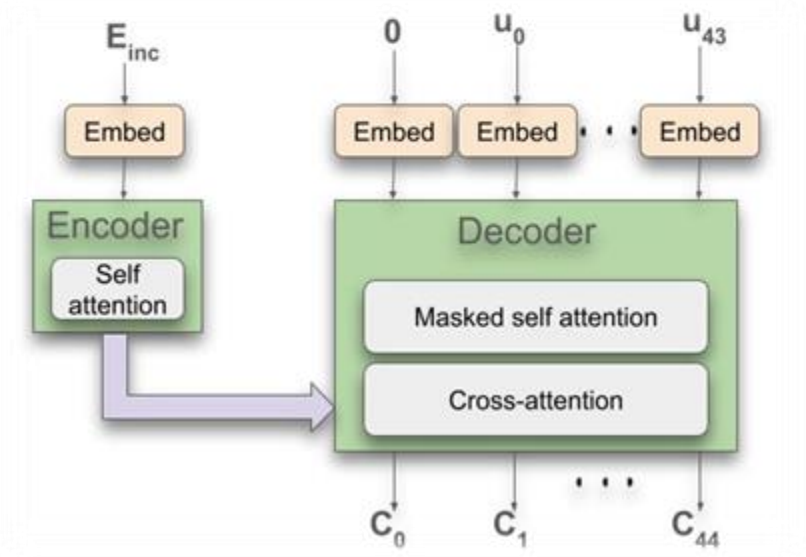
$$u_0 = \frac{\sum_i E_i}{f E_{inc}}, \qquad u_i = \frac{E_i}{\sum_{j \geq i} E_j}, \qquad i = 1,2,\dots,44$$

- E_i : total energy deposited in layer i; f: scale factor.
- Ensures **energy conservation** and keeps $u_i \in [0,1], \forall i > 0$
- Due to detector calibration, u_0 can exceed 1 → rescale with $f = 2.85$.
- Networks are conditioned on $\log(E_{inc})$ after rescaling into $[0,1]$

RQ3: Autoregressive Energy Network



RQ3: Autoregressive Energy Network



$$c_i = \begin{cases} f(E_{inc}), & i = 0 \\ f(u_0, u_1, \dots u_{i-1}, E_{inc}), & i > 0 \end{cases}$$

Where c_i encodes the information necessary to predict the deposited energy of layer i

$$\hat{\epsilon}_{full}(u(t), t, E_{inc}) = [\epsilon_\phi(u_0(t), c_0, t), ; \epsilon_\phi(u_1(t), c_1, t), ; \dots, ; \epsilon_\phi(u_{44}(t), c_{44}, t)]$$

$$\mathcal{L}_\phi = \mathbb{E}_{x_0,t,\epsilon}\left[\left|\epsilon - \epsilon_\phi(x_t, c, t\ )\right|_2^2\right]$$

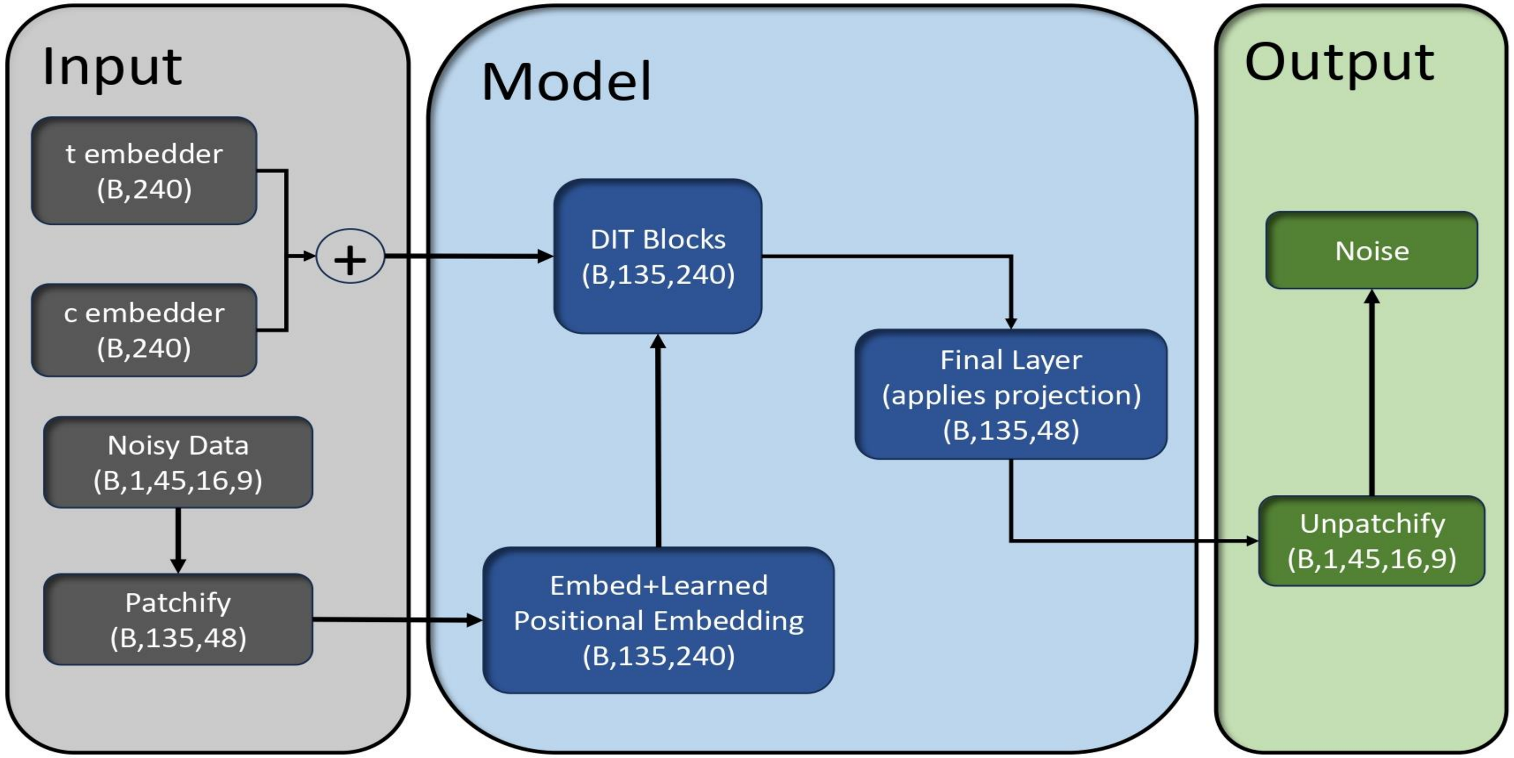
RQ3: Shape Network Preprocessing

- Each **input voxel** x is transformed using a **logit transformation** with small regularization parameter $\alpha = 10^{-6}$

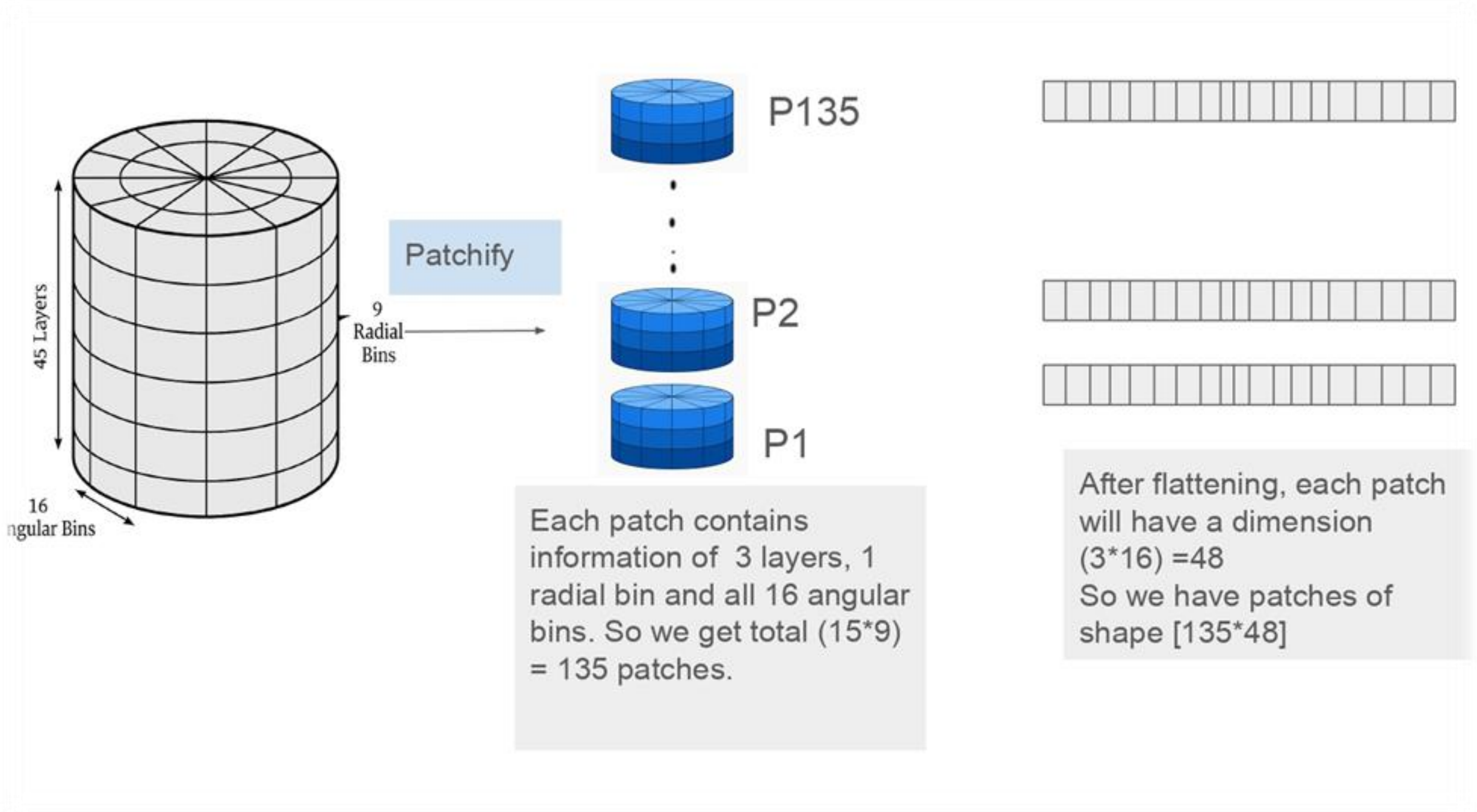
$$x_\alpha = (1 - 2\alpha)x + \alpha, \qquad x' = \log \frac{x_\alpha}{1 - x_\alpha}$$

- This ensures voxel inputs remain in a stable numerical range $[\alpha, 1 - \alpha]$
- Compute **mean and standard deviation** of the training dataset and standardize each voxel feature.
- During postprocessing, **rescale voxel sums per layer** to enforce correct layer-wise normalization.

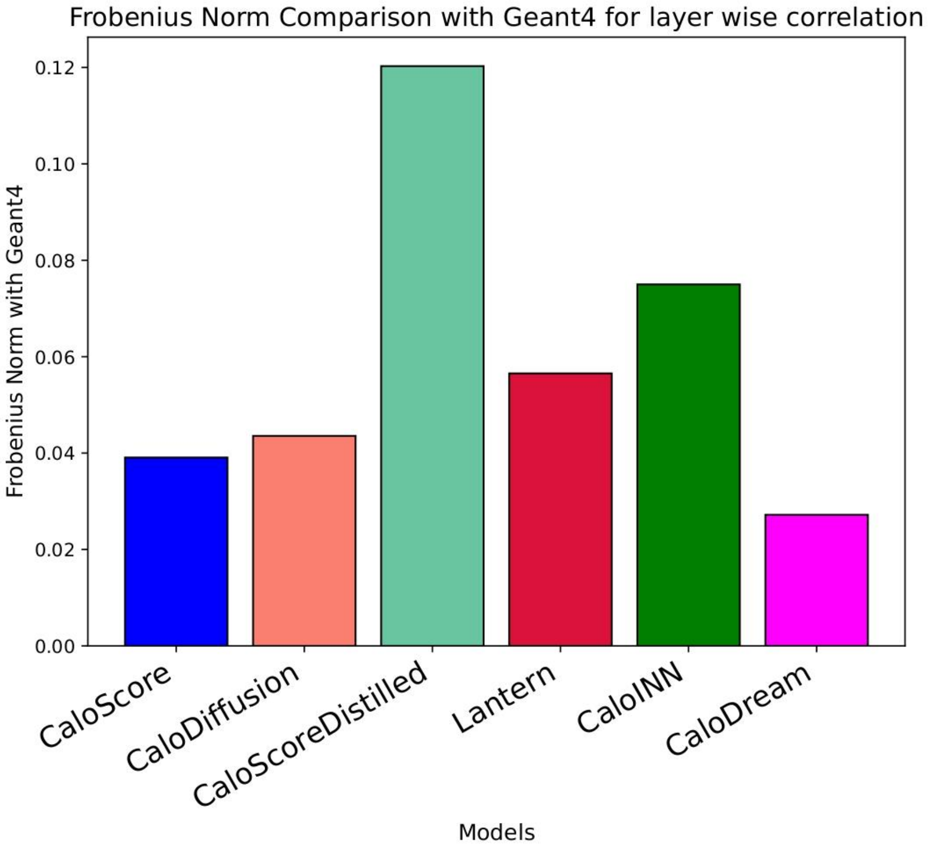
RQ3: Shape Network

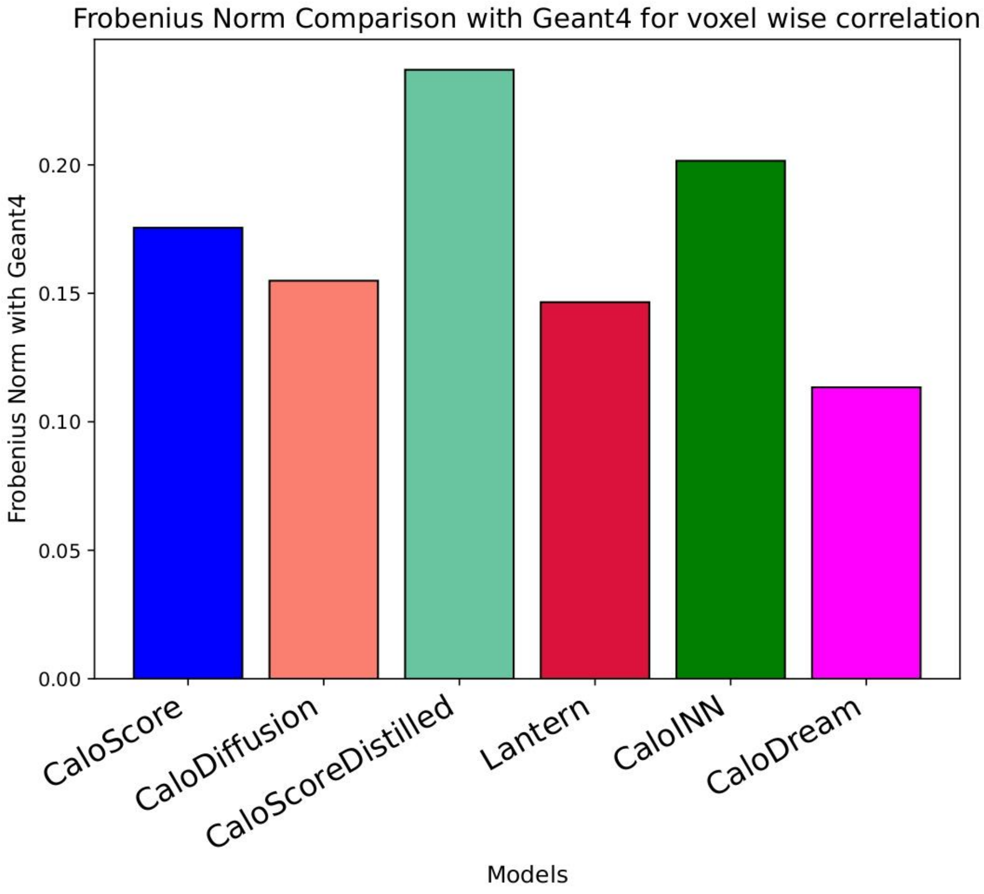


RQ3: Shape Network



RQ3: Results


Frobenius Norm Comparison with Geant4 for layer wise correlation
Frobenius Norm with Geant4
0.12
0.10
0.08
0.06
0.04
0.02
0.00
CaloScore
CaloDiffusion
CaloScoreDistilled
Lantern
CaloINN
CaloDream
Models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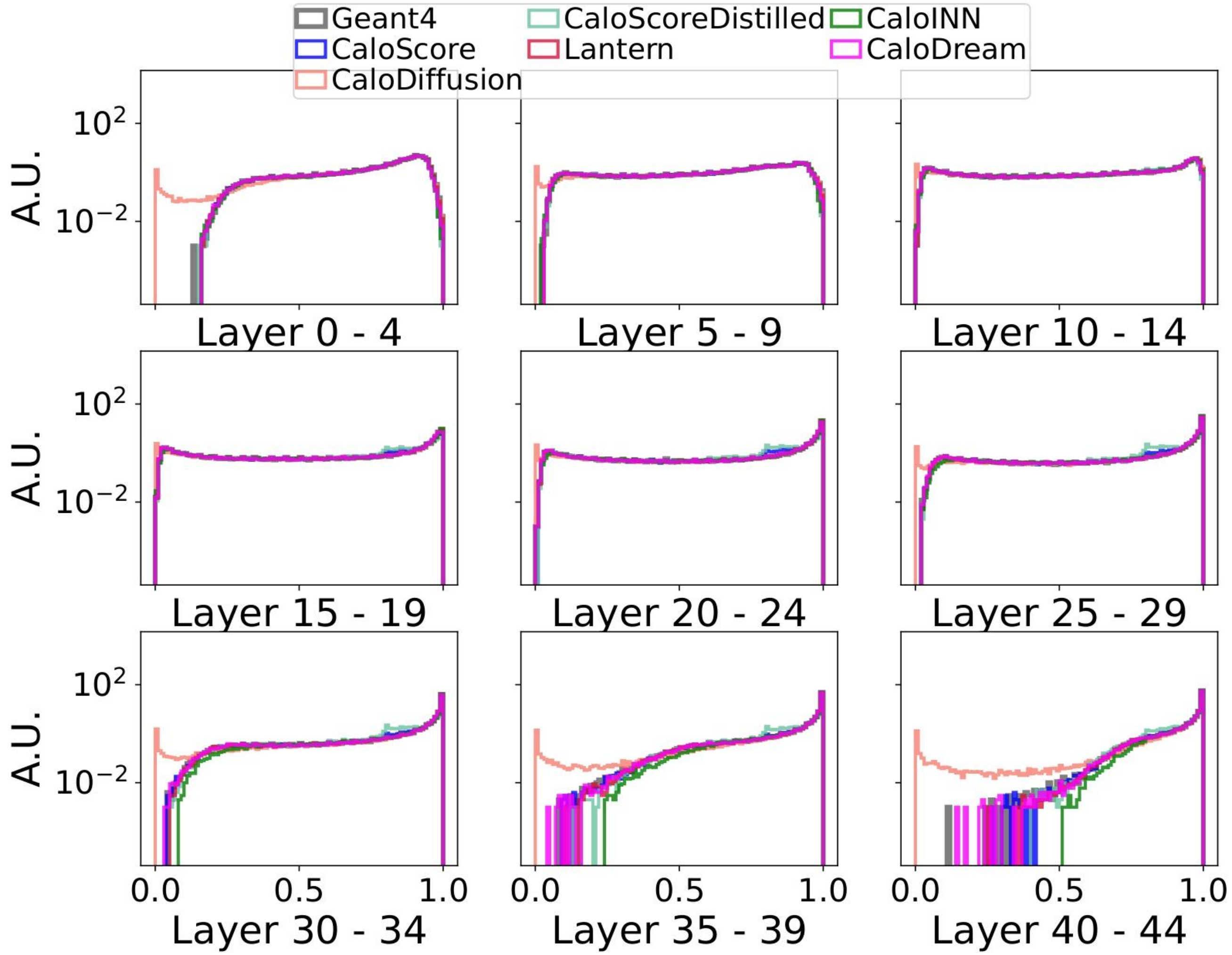
CaloDream
Models

RQ3: Results

- **Lantern preserves sparsity**, unlike other diffusion-based models that struggle to capture empty voxel patterns.
- **Also accurate on key observables** – shower width, center of energy, and energy distributions (angular, radial, longitudinal, and layer-wise).



RQ3: Results

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Discussion – Hypotheses

- **Hypothesis 1**
- *Incorporating domain-informed correlation structures into loss functions or evaluation metrics will improve the quality of generative models and highlight discrepancies that traditional statistical or computer vision metrics fail to capture.*

- **Hypothesis 2**
- *A factorized, two-stage autoregressive diffusion model **'Lantern'** can better capture both layer-wise and voxel-level correlations than existing state-of-the-art generative approaches.*

Discussion – Hypotheses

Hypothesis 1

- *Incorporating domain-informed correlation structures into loss functions or evaluation metrics will improve the quality of generative models and highlight discrepancies that traditional statistical or computer vision metrics fail to capture.*
- Addressed by proposing **CFD**, a physics-aware metric that captures both layer-wise and voxel-wise correlations.
- *Validated as an evaluation tool, but not yet explored as a training objective.*

Hypothesis 2

- *A factorized, two-stage autoregressive diffusion model* ***'Lantern'*** *can better capture both layer-wise and voxel-level correlations than existing state-of-the-art generative approaches.*
- Performs strongly on voxel-wise CFD, but weaker on layer-wise CFD.

Conclusion

- Established a **comprehensive evaluation framework** combining statistical, computer vision, and **physics-informed metrics**, introducing the **Correlation Frobenius Distance (CFD)**.
- **CaloDream (flow matching)** consistently produced the most accurate showers, outperforming diffusion and score-based models.
- Identified limitations: **CaloDiffusion struggled with sparsity**, while other diffusion-based models were competitive.
- Introduced **Lantern**, a two-stage autoregressive diffusion model—achieving strong fidelity with **fewer parameters**, highlighting potential for scalable simulation.

Future Work

- **Correlation-driven training** → incorporate CFD into training objectives for better correlation fidelity
- **Model optimization** → refine Lantern's autoregressive energy predictor & improve modeling of **layer energy correlations** with advanced transformers
- **Cross-geometry generalization** → test Lantern and other models across different calorimeter setups to ensure geometry-agnostic applicability
- **Dataset extension** → apply Lantern to **irregular Dataset 1** and **extensive Dataset 3** for broader validation across particles and energy ranges

Appendix

Evaluation Metrics: Histograms of Physics Observables

Observable	Definition / What it Measures	Usage in Datasets	Interpretation
Layer-wise energy distribution	Energy deposited per layer (summed in groups for large detectors)	DS1: per layer; DS2/DS3: 5-layer groups → 9 plots	Shows shower development along depth of detector
Normalized total energy	Ratio Etot / Einc	All datasets	Tests energy conservation and leakage modeling
Center of energy (η, φ)	Energy-weighted mean position in η or φ	DS1: per layer; DS2/DS3: 5-layer groups → 9 plots	Locates shower core; sensitive to lateral shifts
Shower width (η, φ)	Spread of energy around center of energy	Same grouping as center of energy	Larger = broader; smaller = narrower shower
Sparsity	Fraction of voxels with non-zero energy	Per layer or grouped layers	Tests fine-grained structure; inactive regions
Mean energy distributions	Average energy vs. layer, radial, and angular bins	All datasets	Captures longitudinal and transverse energy flow

Evaluation Metrics: Quantitative Metrics from Histograms

Metric	Definition / Equation	How It's Computed	Interpretation
Separation Power	Normalized difference between two histograms. Equation: $$S^2 = \frac{1}{2}\sum_i \frac{\left(h_{1,i} - h_{2,i}\right)^2}{h_{1,i} + h_{2,i}}$$ $h_{1,i}, h_{2,i}$: i^{th} bin entry of the two histograms being compared	Applied to each observable histogram (Geant4 vs. model).	0 = identical distributions 1 = completely disjoint distributions. Lower = better match.
Earth Mover's Distance (EMD)	Measures minimum "work" to move one distribution into another (a.k.a. Wasserstein-1 distance).	Compare one-dimensional distributions of physics observables.	Lower = closer match between distributions. Sensitive to both sample quality and diversity.

Evaluation Metrics: Physics-Aware Metrics (FPD & KPD)

Metric	Definition / Equation	How It's Computed	Interpretation
Fréchet Physics Distance (FPD)	$FPD(P,Q) = \|\|\mu_P - \mu_Q\|\|^2 + Tr(\Sigma_P + \Sigma_Q - 2\left(\Sigma_P \Sigma_Q\right)^{\frac{1}{2}})$	Compute means (μ) and covariances (Σ) of physics features such as layer energies, shower widths, centers, etc.	Physics version of FID; captures differences in average values and correlations. Lower FPD = better match.
Kernel Physics Distance (KPD)	$KPD(P,Q) = E[k(x,x')] + E[k(y,y')] - 2E[k(x,y)]$	Uses kernel function (e.g. Gaussian) on physics features to compare real vs generated distributions.	Physics version of KID; detects subtle, higher-order discrepancies. Lower KPD = better match.

Evaluation Metrics: Physics-Aware Metrics (FPD)

$$FPD(P,Q) = |\mu_P - \mu_Q|^2 \; + \; \mathrm{Tr}\left(\Sigma_P + \Sigma_Q - 2(\Sigma_P\Sigma_Q)^{1/2}\right)$$

- P: distribution of real physics observables
- Q: distribution of generated observables
- μ_P, μ_Q: feature means of P and Q
- Σ_P, Σ_Q: covariance matrices of P and Q

Analogous to Fréchet Inception Distance (FID) in computer vision, but computed on physics features instead of image embeddings.

Smaller FPD = generated distribution closer to real data.

Evaluation Metrics: Physics-Aware Metrics (KPD)

$$KPD(P,Q) = \mathbb{E}_{x,x' \sim P}[k(x,x')] + \mathbb{E}_{y,y' \sim Q}[k(y,y')] - 2\,\mathbb{E}_{x \sim P,\, y \sim Q}[k(x,y)]$$

- x, x': samples from real distribution P
- y, y': samples from generated distribution Q
- $k(\cdot,\ \cdot)$: kernel function measuring similarity
  - Common choice: Gaussian RBF,

    $$k(x,y) = \exp\left(-\frac{|x-y|^2}{2\sigma^2}\right)$$
  - σ: bandwidth parameter controlling sensitivity of similarity
  - $k(x,y) \approx 1$ if $x \approx y$, small if far apart

If $P = Q$, then $KPD = 0$ (perfect match). Smaller values = closer distributions.

Evaluation Metrics: Manifold-based Metrics (Precision)

- Measures quality of generated samples: fraction of synthetic samples that fall inside the real data manifold.
- Precision = $(\frac{1}{|Y|})\ \Sigma_{\mathrm{y} \in Y}\ \mathbb{1}[y\ \in\ M_k(X)\ ]$
- X: real samples, Y: generated samples. $M_k(X)$: k-NN manifold of real data. Counts how many generated samples lie inside real neighborhoods.
- <u>Motivation:</u> Introduced to detect mode inventing (bad quality) which single metrics like FID cannot capture.
- <u>Limitations:</u> Can be inflated by outliers (mode inventing). Sensitive to neighborhood size k.

Evaluation Metrics: Manifold-based Metrics (Recall)

- . Measures diversity of generated samples: fraction of real samples that lie inside the generated data manifold.
- . Recall = $\left(\frac{1}{|X|}\right) \Sigma_{x \in X} \mathbb{1}[x \in M_k(Y)]$
- X: real samples, Y: generated samples. $M_k(Y)$: k-NN manifold of generated data. Counts how many real samples are covered by generated neighborhoods.
- <u>Motivation:</u> Introduced to detect mode dropping (missing diversity), complementing precision.
- <u>Limitations:</u> May overestimate if generated distribution is overly broad. Sensitive to k.

Evaluation Metrics: Manifold-based Metrics (Density)

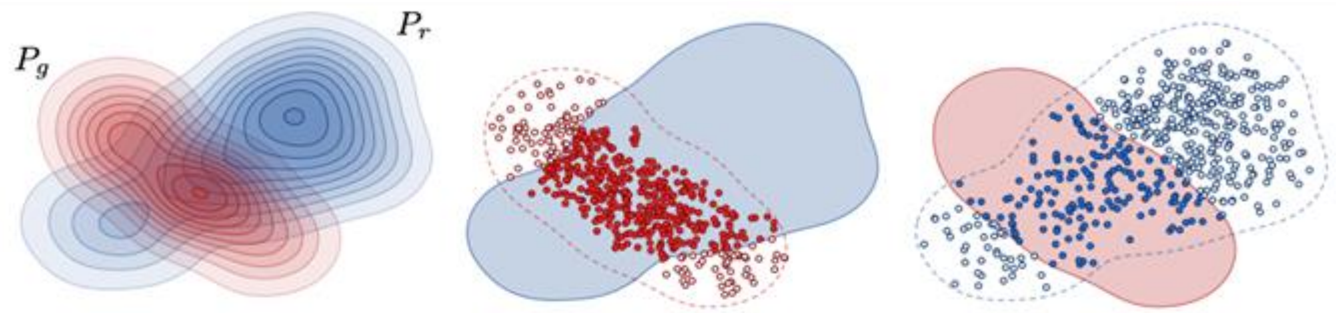
- Measures local concentration of generated samples in real neighborhoods (mode fidelity).
- Density = $\left(\frac{1}{k|Y|}\right) \Sigma_{y \in Y} \Sigma_{\mathrm{x} \in X} \mathbb{1}[y \in B(x, d_{k,X}(x))]$
- $B(x, d_{k,X}(x))$: neighborhood around real sample $x, d_{k,X}(x)$ distance to k-th neighbor of x in real data. Counts average real neighborhoods hit by each generated sample.
- <u>Motivation:</u> Refines precision by preventing overestimation of quality when generated samples cluster too tightly.
- <u>Limitations:</u> Not bounded by 1. Sensitive to sample imbalance and choice of k.

Evaluation Metrics: Manifold-based Metrics (Coverage)

- •Measures coverage of real data: fraction of real neighborhoods covered by at least one generated sample.
- Coverage = $\left(\frac{1}{|X|}\right) \Sigma_{x \in X} \mathbb{1}[\exists\ y\ \in\ Y \mid y\ \in\ B(x, d_k, X(x))\ ]$
- X: real samples, Y: generated samples. B(x,r): neighborhood around real sample x. Checks if each real region has at least one generated sample.
- Motivation: Refines recall by avoiding inflated scores when generated distribution is too broad.
- Limitations: Binary measure: ignores how many generated samples fall in each region. Sensitive to k.

Evaluation Metrics: Manifold-based Metrics (PRDC)

	Recall Low	Recall High
Precision Low	Mode Dropping	Mode Inventing
Precision High	Mode Collapse	Good Model



(a) Example distributions (b) Precision (c) Recall

Figure 1: Definition of precision and recall for distributions [25]. (a) Denote the distribution of real images with P_r (blue) and the distribution of generated images with P_g (red). (b) Precision is the probability that a random image from P_g falls within the support of P_r. (c) Recall is the probability that a random image from P_r falls within the support of P_g.

Ref: [30]

Evaluation Metrics: Manifold-based Metrics (PRDC)

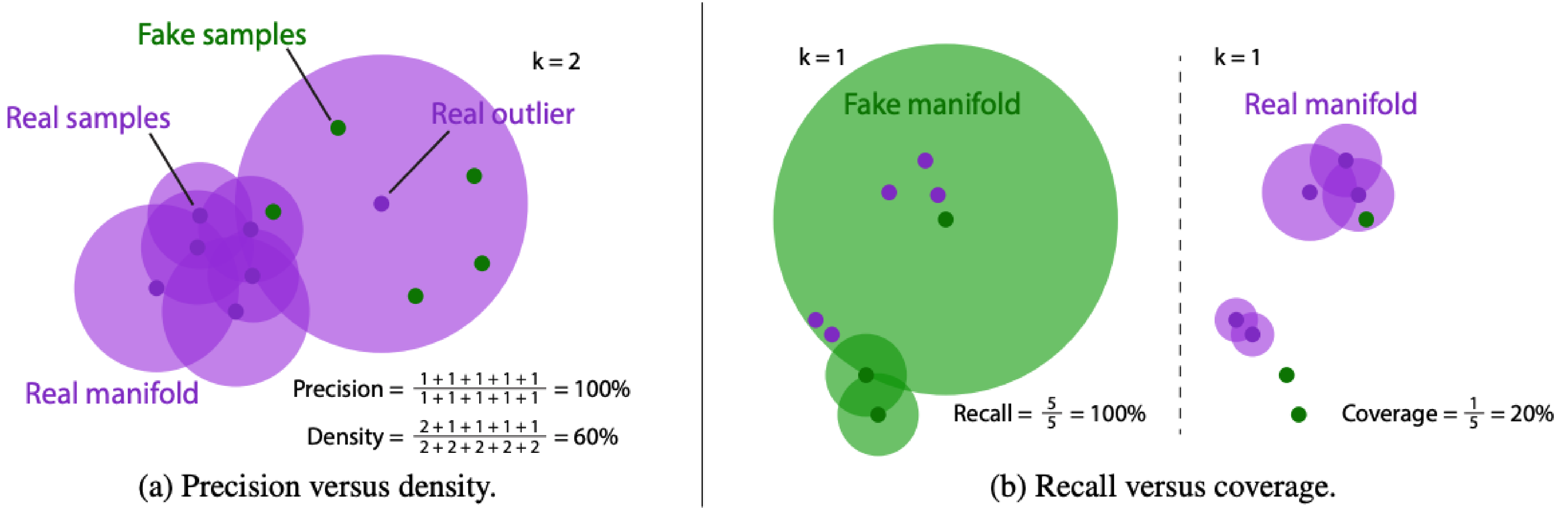
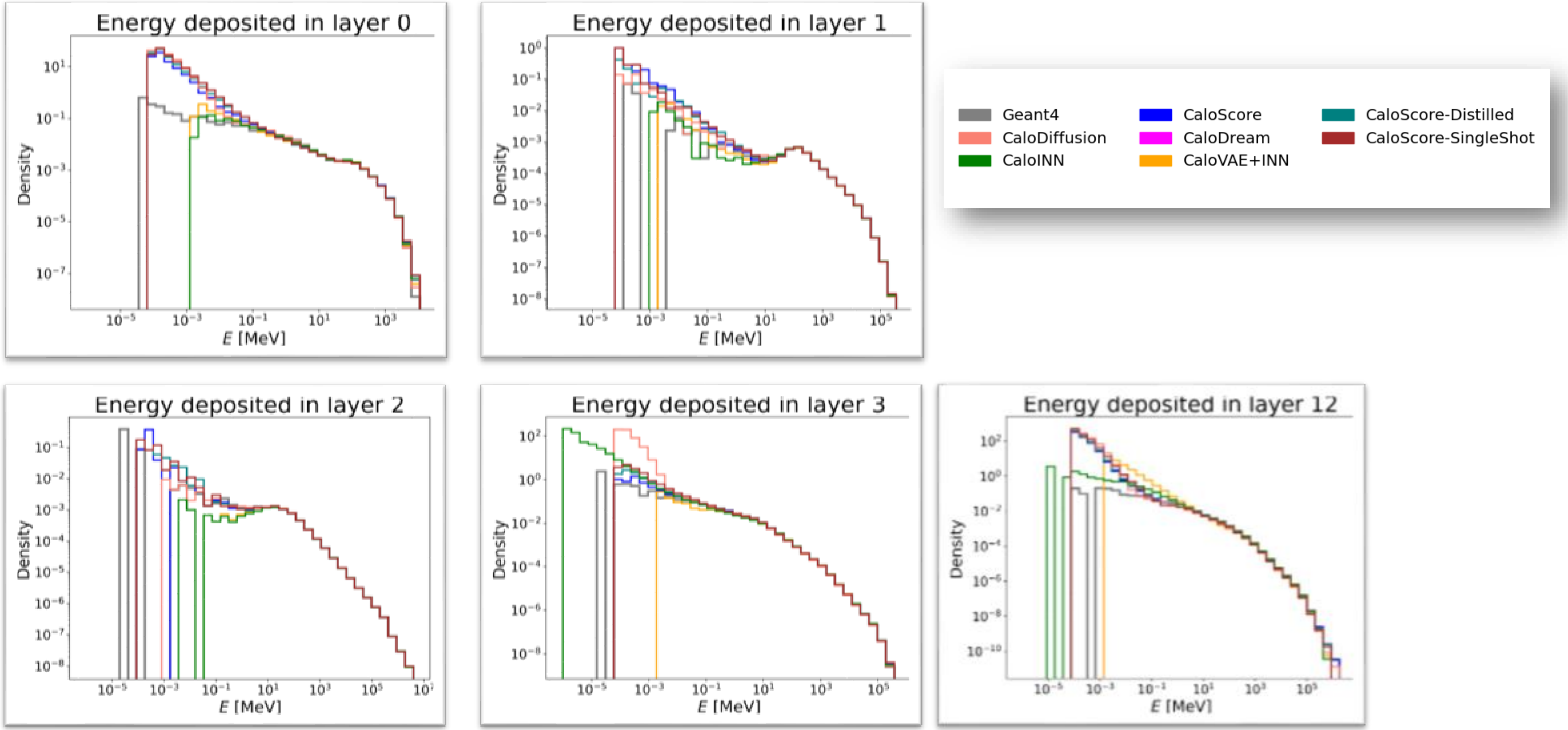


Figure 1. **Overview of metrics.** Two example scenarios for illustrating the advantage of density over precision and coverage over recall. Note that for recall versus coverage figure, the real and fake samples are identical across left and right.

Ref: [27]

RQ2: Results (Dataset 1: Photons)



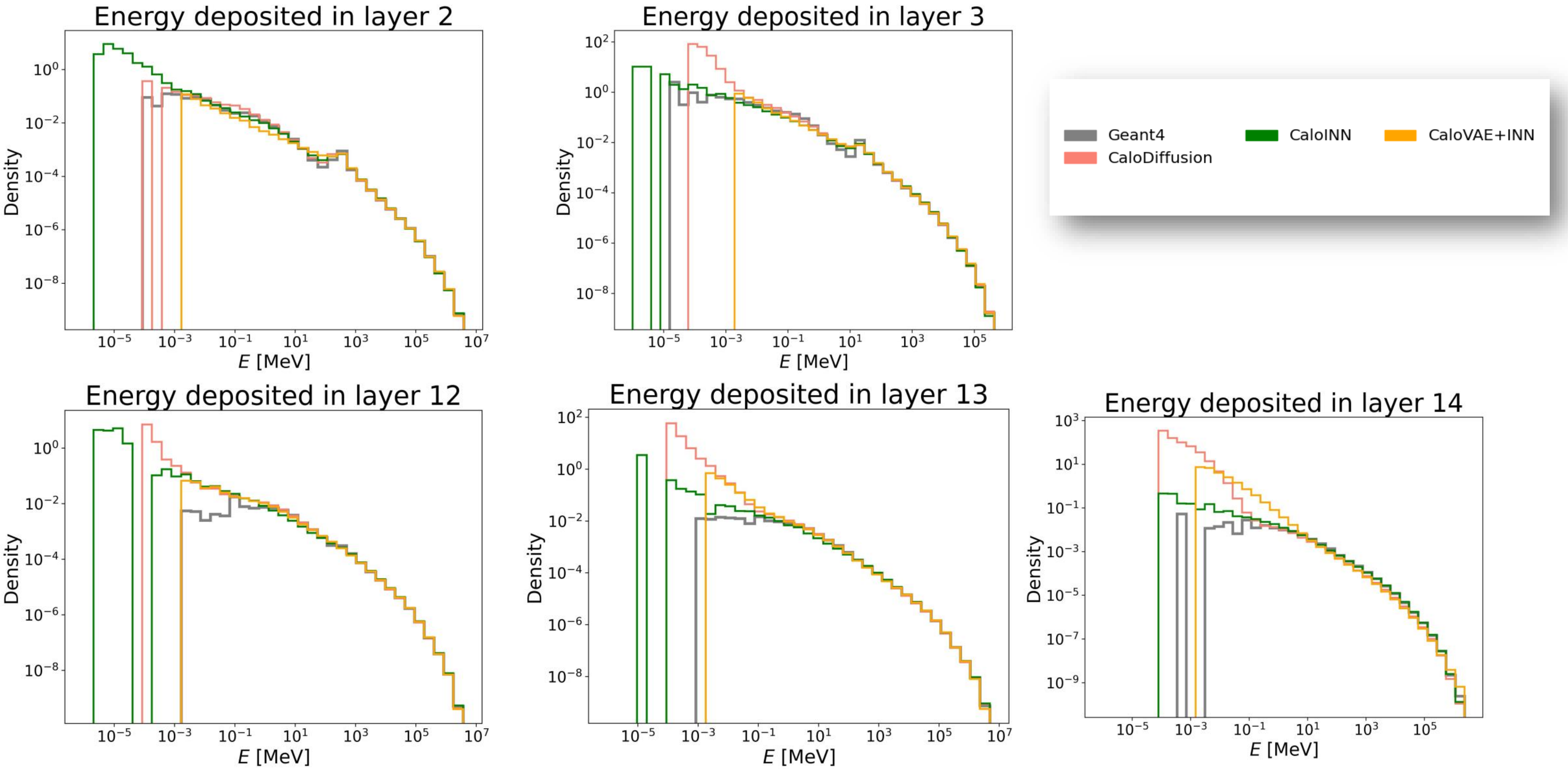
RQ2: Results (Dataset 1: Photons)

Model	FPD/KPD (↓)	Classifier Test (~0.5) (Low and High level features)	Precision, Recall, Density, Coverage (PRDC) (↑)	# of Parameters (↓)
CaloDiffusion	low	0.5-0.6	balanced	~0.52M
CaloScore	low	0.5-0.6	balanced	~2.25M
CaloINN	high	>0.8	Low R/C	~18.8M

RQ2: Results (Dataset 1: Photons)

Model	Accuracy	Generation Speed	Memory Requirement
CaloDiffusion	High	Moderate	Least
CaloScore	High	Slowest	Moderate
CaloINN	Low	Fastest	Highest

RQ2: Results (Dataset 1: Pions)



RQ2: Results (Dataset 1: Pions)

Model	FPD/KPD ( )	Classifier Test (~0.5) (Low and High level features)	Precision, Recall, Density, Coverage (PRDC) ( )	# of Parameters ( )
CaloDiffusion	low	≈ 0.68/0.65	balanced	~0.52M
CaloINN	high	≈0.78/0.73	low R/C	~26.5M
CaloVAE+INN	high	≈0.85/0.92	imbalanced	~9.1M

RQ2: Results (Dataset 1: Pions)

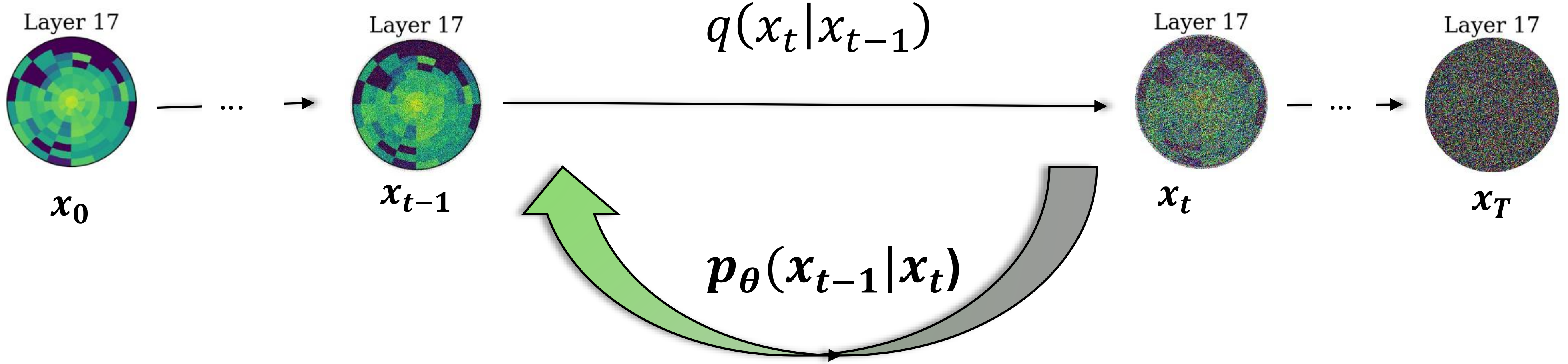
Model	Accuracy	Generation Speed	Memory Requirement
CaloDiffusion	High	Slowest	Least
CaloINN	Low	Fastest in GPU	Highest
CaloVAE+INN	Low	Fastest in CPU	High

RQ3: Dataset 2 Simulation Setup & Voxelization

- **Dataset 2 (DS2)** from the **CaloChallenge 2022** is used for benchmarking.
- Contains **200k GEANT4-simulated electron showers**: 100k for training/validation and 100k for testing.
- Showers are generated over a log-uniform incident energy range: E_{inc} **= 10^3 … 10^6 MeV**
- Detector has a cylindrical geometry with alternating absorber/active layers; voxelization merges them into **45 concentric cylindrical layers.**
- DS2 consists of **6480 voxels (144 per layer: 16 angular × 9 radial bins)**.
- A **15.15 keV** threshold is applied on energy deposits. While unrealistic for practical detectors, it provides a useful challenge for high-dimensional generative models.

Diffusion Model

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \mu_t = \sqrt{1-\beta_t}\,x_{t-1}, \Sigma_t = \beta_t \boldsymbol{I})$$



Diffusion Model – Forward Process

- Forward process gradually perturbs clean shower data x_0 by adding Gaussian noise over T steps.

- Each step defined as: $q(x_t|x_0) = \mathcal{N}\left(\sqrt{\overline{\alpha_t}}x_0, (1 - \overline{\alpha_t}\boldsymbol{I})\right)$

- Closed form of noisy data: $x_t = \sqrt{\overline{\alpha_t}}x_0 + \left(\sqrt{1 - \overline{\alpha_t}}\right)\epsilon, \epsilon \sim \mathcal{N}(0,1)$
- $\alpha_t = 1 - \beta_t$, where $\{\beta_t\}_{t=1}^{T}$ is the variance schedule.
- $\overline{\alpha_t} = \prod_{i=1}^{t} \alpha_i$
- **Intuition:** gradually destroys structure while retaining statistical traces.

Diffusion Model – Reverse Process

- Goal: Learn reverse process $p_\theta(x_{t-1}|x_t)$ to reconstruct clean showers.

- Reverse update:

$$x_{\mathrm{t-1}} = \frac{1}{\sqrt{\alpha_{\mathrm{t}}}}\left(x_{\mathrm{t}} - \frac{1-\alpha_{\mathrm{t}}}{\sqrt{(1-\bar{\alpha}_{\mathrm{t}})}}\,\epsilon_\theta(x_t, t, c)\right) + \sigma_t z$$

where $z \sim \mathcal{N}(0,1)$

- Condition c: incident energy, layer embeddings, or autoregressive context.
- Training objective: predict added noise ε with MSE loss:

$$\mathcal{L}_{DDPM} = \mathbb{E}_{x_0,t,\epsilon}\left[\left|\left|\epsilon - \epsilon_\phi(x_t, c, t\,)\right|\right|_2^2\right]$$

- Iteratively applying reverse steps → generates high-fidelity calorimeter showers.

Diffusion Model : Sampling

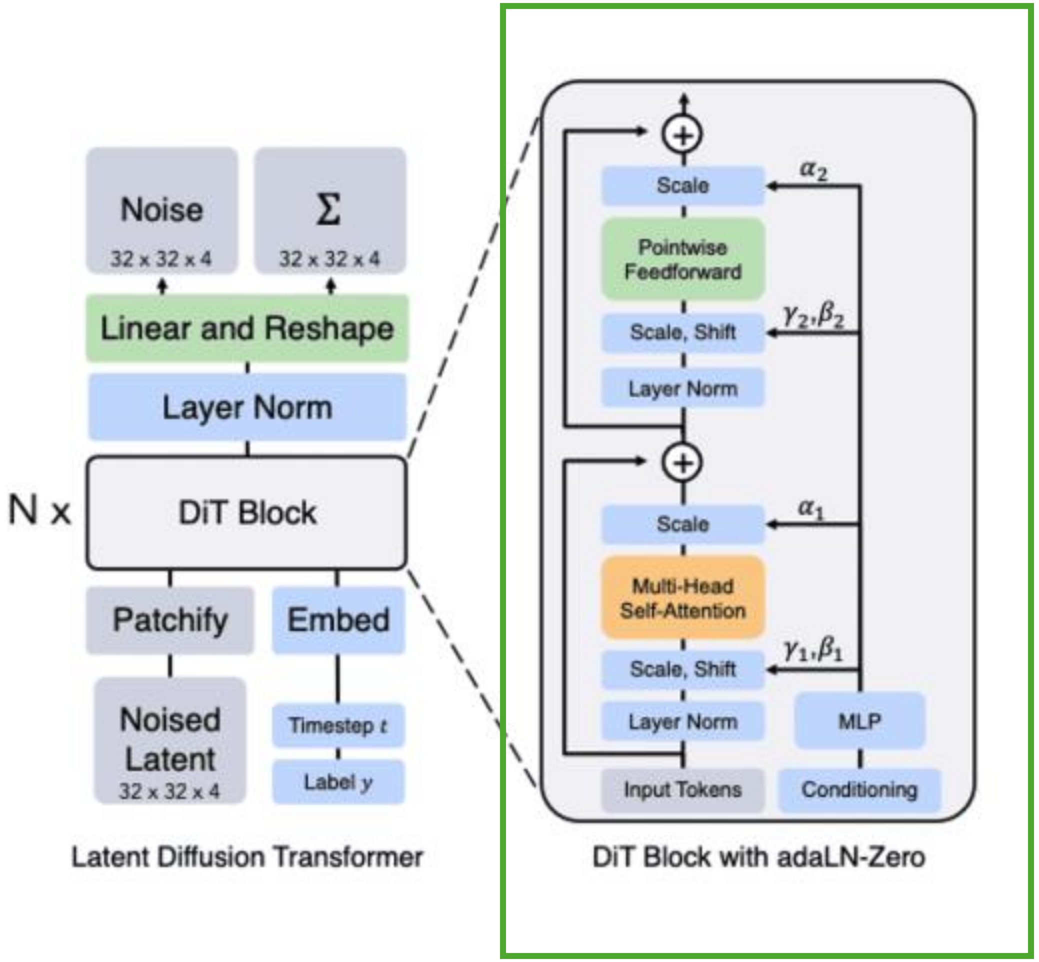
1. Start from pure noise: $x_T \sim \mathcal{N}(0, \boldsymbol{I})$
2. Define $\alpha_t = 1 - \beta_t,\ \overline{\alpha_t} = \prod_{s=1}^{t} \alpha_s$
3. Noise prediction: $\epsilon_\theta(x_t, t, c)$ (Condition: e.g., E_{inc}, Layer Context)
4. Reverse Update (DDPM, noise-prediction form):

$$x_{t-1} = \frac{1}{\sqrt{\alpha_\mathrm{t}}}\left(x_\mathrm{t} - \frac{\beta_t}{\sqrt{(1-\bar{\alpha}_\mathrm{t})}}\ \epsilon_\theta(x_t, t, c)\right) + \sigma_t z\,, \qquad z \sim \mathcal{N}(0, \boldsymbol{I})$$

Where $\sigma_t^2 = \beta_t \frac{1-\overline{\alpha_{t-1}}}{1-\overline{\alpha_t}}$

Scalable Diffusion Models with Transformers[29]

- Traditional diffusion models (UNet-based) are **convolution-heavy** → good for images but less flexible for non-grid physics data
- Transformers provide **global attention**, capturing long-range correlations across spatial dimensions.
- Calorimeter showers: inherently **high-dimensional and correlated across layers & voxels** → ideal use case for DiT



Flow Matching: The idea

- Goal: transform a simple source distribution (e.g., Gaussian) into a complex target distribution (data).
- Learn a **vector field** $u_t(x)$ that tells us how points move over time.
- By integrating this ODE:

$$\frac{dx}{dt} = u_t(x),$$

we evolve samples from noise to data.

Challenge in Flow Matching

- Directly estimating $u_t(x)$ for the whole distribution is **intractable**.
- Requires marginalizing over all possible paths from noise → data.
- We need a **tractable training objective**.

Conditional Flow Matching (CFM)

- Instead of learning the global flow, we define **conditional paths** between pairs of samples.
- Pick $x_0 \sim p_0$ (source) and $x_1 \sim p_1$ (data).
- Define a simple interpolation:

$$x_t = (1 - t)x_0 + tx_1.$$

- The conditional vector field along this path is:

$$u_t(x \mid x_1) = \frac{d}{dt}x_t = x_1 - x_0.$$

- Train the model to approximate these conditional vector fields.

Why CFM Works

- Conditional flows are **analytically tractable** (linear interpolation, Gaussian bridges, etc.).
- The expectation of conditional vector fields recovers the true global vector field:

$$\mathbb{E}_{x_0,x_1}[u_t(x \mid x_1)] = u_t(x).$$

- Minimizing the **CFM loss** gives the same gradient as full flow matching, but is much easier to compute.
- End result: efficient training, correct objective.

Comparison of Diffusion and CFM

Aspect	**Diffusion Models**	**Conditional Flow Matching (CFM)**
Training Objective	Reverse a stochastic noising process; model learns to predict added noise (ε).	Learn a deterministic flow (ODE) by matching conditional vector fields.
Trajectories	Stochastic: randomness is injected at every denoising step.	Deterministic: samples evolve smoothly along interpolation paths.
Probability Modeling	Convergence guaranteed by forward–reverse diffusion process.	Averaging conditional flows recovers the true global flow field.
Computation	Needs many denoising steps (50–1000) for high quality results.	Can use fewer steps in principle, since it integrates an ODE (solver dependent).

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